

Encryption Technologies

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Cryptography

- "Cryptography" comes from the Greek words kryptos, meaning "concealed, hidden, veiled, secret, or mysterious," and graphia, meaning "writing"; thus, cryptography is "the art of secret writing."
- Cryptography is the practice of concealing information by converting plaintext (readable format) into ciphertext (unreadable format) using a key or encryption scheme.
- It is the process of converting data into a scrambled code that is encrypted and sent across a private or public network.
- **Encryption** is the method by which information is converted into secret code that hides the information's true meaning
- Encrypted messages can, at times, be decrypted by **cryptanalysis** (code breaking), even though modern encryption techniques are virtually unbreakable.

Objectives of Cryptography

- **Confidentiality**: Assurance that the information is accessible only to those authorized to access it.
- **Integrity:** Trustworthiness of data or resources in terms of preventing improper and unauthorized changes.
- **Authentication:** Assurance that the communication, document, or data is genuine.
- Nonrepudiation: Guarantee that the sender of a message cannot later deny having sent the message and that the recipient cannot deny having received the message.

Cryptography Terminology

- **Plain text** – Message in its original form (readable text)
- **Cipher text** – Message in Coded Form
- **Encryption** – Transforming Plain text to Cipher text
- **Decryption** – transforming cipher text to plain text
- **Encryption algorithm** – Method used for encryption

Cryptography Process

- The cryptography process involves transforming a Plaintext **(readable format)** into a cipher text using encryption algorithms.
- On reaching the destination, is decrypt cipher text into readable plaintext.



Types of Cryptography

- Cryptography is categorized into two types according to the number of keys employed for encryption and decryption:
  - Symmetric encryption
  - Asymmetric Encryption

Symmetric encryption

- Symmetric encryption use the same pre-shared key (secret key) to encrypt and decrypt data.
- The sender uses a key to encrypt the plaintext and sends the resultant cipher text to the recipient, who uses the same key (used for encryption) to decrypt the cipher text into plaintext.
- Symmetric encryption is also known as secret-key cryptography, as it uses only one secret key to encrypt and decrypt the data.

Symmetric encryption



Symmetric encryption

- Symmetric encryption comes in two forms, these are:
  - Block cipher
  - Stream ciphers

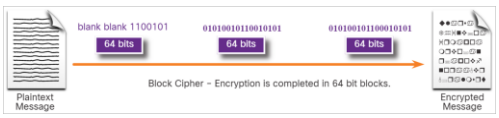
Symmetric Encryption (Contd.)

Block Ciphers

- A block cipher is a method of encrypting data in blocks to produce cipher text using a cryptographic key and algorithm.
- The block cipher processes fixed-size blocks simultaneously
- Block ciphers transform a fixed-length block of plaintext into a common block of ciphertext of 64 or 128 bits.
- Common block ciphers include Data Encryption Standard(DES) with a 64-bit block size and Advanced Encryption Standard (AES) with a 128-bit block size.

Symmetric encryption

Block Ciphers

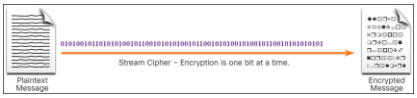


Symmetric Encryption (Contd.)

Stream Ciphers

- Stream ciphers encrypt plaintext one byte or one bit at a time.
- Stream ciphers are basically a block cipher with a block size of one byte or bit.
- Stream ciphers are typically faster than block ciphers because data is continuously encrypted.
- Examples include RC4 and A5 which is used to encrypt Global System for Mobile communication(GSM)cell phone communications.

Symmetric encryption



Uses of symmetric encryption

1. **Data at rest:** Symmetric cryptography is also widely used to secure data stored in hard drives, archives, computers, flash drives, and laptops.
2. **Banking sector** – used Symmetric encryption to encrypt bulk data within the banking sector
3. **Messaging apps** like WhatsApp and signal uses end to end encryption to safeguard the privacy and confidentiality of communications on the platforms and also in authenticating users.

Symmetric encryption

- This type of cryptography works well when you are communicating with only a few people.
- Because the sender and receiver must share the key before sending any messages, this technique is of limited use for the Internet, where individuals who have not had prior contact frequently require a secure means of communication.
- The solution to this problem is asymmetric encryption (public-key cryptography).

Asymmetric Encryption

- An asymmetric-key system is an encryption method that uses a key pair comprising a public key available to anyone and a private key held only by the key owner, which helps to provide confidentiality, integrity, authentication, and nonrepudiation in data management.
- The concept of asymmetric encryption (also known as public-key cryptography) was introduced to solve key-management problems.
- Asymmetric encryption involves both a public key and a private key.

Asymmetric Encryption

Asymmetric encryption uses the following sequence to send a message:

1. An individual finds the public key of the person he or she wants to contact in a directory.
2. This public key is used to encrypt a message that is then sent to the intended recipient.
3. The receiver uses the private key to decrypt the message and reads it.

Asymmetric Encryption

- No one but the holder of the private key can decrypt a message encrypted with the corresponding public key.
- This increases the security of the information because all communications involve only public keys; the message sender never transmits or shares the private keys.
- The sender must link public keys with usernames in a secure manner to ensure that individuals claiming to be the intended recipient do not intercept the information.



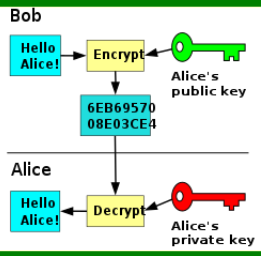
Asymmetric encryption

- Common examples of asymmetric encryption algorithms
  - Digital Signature Standard (DSS) and Digital Signature Algorithm (DSA)
  - Diffie-Hellman (DH)
  - Elliptic curve techniques

Asymmetric Encryption

- Asymmetric algorithms are used to provide confidentiality without pre-sharing a password.
- The confidentiality objective of asymmetric algorithms is initiated when the encryption process is started with the public key.
- The process can be summarized using the formula: **Public Key (Encrypt) + Private Key (Decrypt) = Confidentiality**
- When the public key is used to encrypt data, the private key must be used to decrypt data.
- Only one host has the private key; therefore, confidentiality is achieved.

Asymmetric Encryption



Discussion

Group 1: What do you think are the Strengths and weaknesses of asymmetric encryption?

Group 2 : What do you think are the Strengths and weaknesses of symmetric encryption?

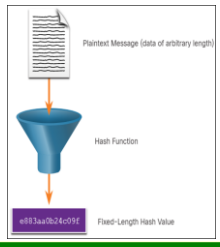
Strengths and weaknesses of asymmetric and symmetric encryption

- Take note of the weaknesses and strengths discussed

Hash Functions

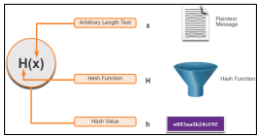
- Hashing is based on a one-way mathematical function that is relatively easy to compute, but significantly harder to reverse.
- Hashes are used to verify and ensure data integrity.
- A hash function takes a variable block of binary data, called the message, and produces a fixed-length, condensed representation, called the hash.
- The resulting hash is also sometimes called the message digest, digest, or digital fingerprint.
- With hash functions, it is computationally infeasible for two different sets of data to come up with the same hash output.
- Every time the data is changed or altered, the hash value also changes – used in checking integrity of data

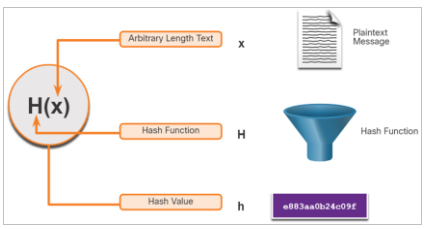
How Hashing Works



Hash Operation

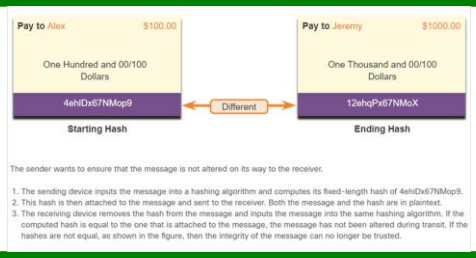
- Mathematically, the equation ***h= H(x)*** is used to explain how a hash algorithm operates.
- As shown in the figure, a hash function H takes an input x and returns a fixed-size string hash value h.
- A cryptographic hash function should have the following properties:
  - The input can be any length.
  - The output has a fixed length.
  - H(x) is relatively easy to compute for given x.
  - H(x) is one way and not reversible.
  - H(x) is collision free, meaning that two different input values will result in different hash values.





Hash Functions

- Hash functions are used to ensure the integrity of a message either accidentally or intentionally.
- In the figure, the sender is sending a $100 money transfer to Alex. The sender wants to ensure that the message is not altered on its way to the receiver.



MD5 and SHA

There are four well-known hash functions:

- **MD5 with 128-bit digest** - A one-way function that produces a 128-bit hashed message. MD5 is a legacy algorithm.
- **SHA-1 -** Very similar to the MD5 hash functions. SHA-1 creates a 160-bit hashed message and is slightly slower than MD5.
- **SHA-2 -** If you are using SHA-2, then SHA-256, SHA-384, and SHA-512 algorithms should be used.
  - **SHA-3 -** Next-generation algorithms and should be used whenever possible.

MD5 and SHA

- While hashing can be used to detect accidental changes, it cannot be used to guard against deliberate changes that are made by a threat actor.
- There is no unique identifying information from the sender in the hashing procedure.
- This means that anyone can compute a hash for any data, as long as they have the correct hash function.
- Therefore, hashing is vulnerable to man-in-the-middle attacks and does not provide security to transmitted data. To provide integrity and origin authentication, something more is required.

Characteristics of a Hashing algorithm

- A hashing algorithm is considered secure if it has these characteristics:
  - **Fixed size**. A digest of a short set of data should produce the same size as a digest of a long set of data. For example, a digest of the single letter a is 86be7afa339d0fc7cfc785e72f578d33, while a digest of 1 million occurrences of the letter a is 4a7f5723f954eba1216c9d8f6320431f, the same length.
  - **Unique.** Two different sets of data cannot produce the same digest, which is known as a collision. Changing a single letter in one data set should produce an entirely different digest. For example, a digest of Sunday is 0d716e73a2a7910bd4ae63407056d79b while a digest of sunday (lowercase s) is 3464eb71bd7a4377967a30da798a1b54.
  - **Original.** It should be impossible to produce a data set that has a desired or predefined hash.
  - **Secure.** The resulting hash cannot be reversed in order to determine the original plaintext.

Applications of Hash Functions

1. **Password Storage**
   - Hash functions provide protection to password storage.
   - Instead of storing password in clear, mostly all logon processes store the hash values of passwords in the file.
   - Example: Microsoft stores password as hashes, most websites also stores password as hashes.
2. **Data Integrity Check**
   - Data integrity check is a most common application of the hash functions. It is used to generate the checksums on data files.
   - This application provides assurance to the user about correctness of the data.
   - Example: sites that run downloads

Encryption vs Hashing

What are the differences between Hashing and Encryption

LAB 1 – To do at home

Creating Hashes in Windows
Activity(15 Min)

Simple Cryptography – Caesar Cipher

- Caesar Cipher is the earliest ciphers developed by Caeser
- The original one, each letter in a message is substituted by another that is 3 letters away i.e. a becomes D
- Also known as shift cipher
- For Example : Attack at Dawn

Caesar Cipher

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Diffie-Hellman

- Diffie-Hellman key exchange is a protocol that two people can use over a shared network connection to share a key they want to use for encryption i.e in symmetric key encryption.
- How do you share key encryption with someone before you both know the key to use for encryption – Use differ hellman.

Video

- Video explaining how diffie Hellman works

Digital Signatures

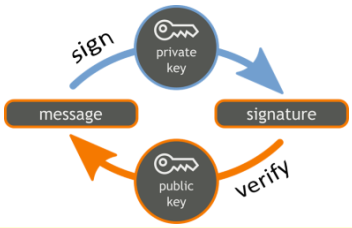
- A digital signature is a small amount of data that was created using some private key, and there is a public key that can be used to verify that the signature was really generated using the corresponding private key.
- Originates from asymmetric cryptography
- Digital signature is a cryptographic value that is calculated from the data and a secret key known only by the signer.
- The algorithm used to generate the signature must be such that without knowing the private key it is not possible to create a signature that would verify as valid.
- Digital signatures are used to verify that a message really comes from the claimed sender (assuming only the sender knows the private key corresponding to the public key). This is called (data origin) authentication.

Digital Signature

- They can also be used to timestamp documents: a trusted party signs the document and its timestamp with his/her private key, thus testifying that the document existed at the stated time.
- Digital signatures can also be used to certify that a public key belongs to a particular entity.

Digital Signatures: How they work

A message is signed by a private key and the signature is verified by the corresponding public key



Digital Signatures: How they work

Messages are signed by the sender using a private key (signing key).

Typically the input message is hashed and then the signature is calculated by the signing algorithm.

Most signature algorithms perform some calculation with the message hash + the signing key in a way that the result cannot be calculated without the signing key.

- The result from message signing is the digital signature (one or more integers):
- **signMsg(msg, privKey)**

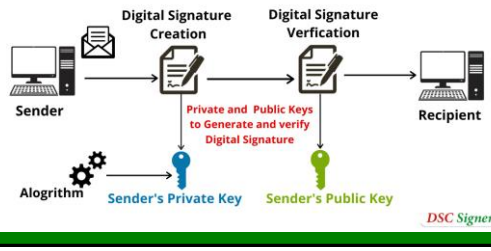
Message signatures are verified by the corresponding public key (verification key).

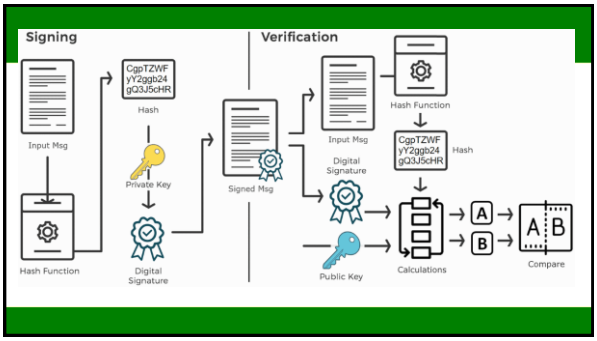
Typically the signed message is hashed and some calculation is performed by the signature algorithm using the message hash + the public key.

The result from signing is a boolean value (valid or invalid signature):

verifyMsgSignature(msg, signature, pubKey) valid / invalid

How Digital Signature Works





Importance of Digital signatures

1. **Message authentication** − When the verifier validates the digital signature using public key of a sender, he is assured that signature has been created only by sender who possess the corresponding secret private key and no one else.
2. **Data Integrity** − In case an attacker has access to the data and modifies it, the digital signature verification at receiver end fails. The hash of modified data and the output provided by the verification algorithm will not match. Hence, receiver can safely deny the message assuming that data integrity has been breached.
3. **Non-repudiation** − Since it is assumed that only the signer has the knowledge of the signature key, he can only create unique signature on a given data. Thus the receiver can present data and the digital signature to a third party as evidence if any dispute arises in the future.

Implementing encryption in networks

- The implementation of encryption in packet-switched networks must ensure that essential addressing information can be accessed by the relevant network devices such as switches, bridges and routers.
- Encryption is broadly termed link layer encryption or end-to-end encryption depending on whether it is applied and re-applied at each end of each link in a communication path, or whether it is applied over the whole path between end systems.
- It is useful to identify the various implementations of encryption with the appropriate OSI layer
- End-to-end encryption is implemented at or above layer 3, the network layer of the OSI reference model, while link layer encryption is applied at the data link and physical layers.
- When encryption is applied at the transport or network layers, end-to-end refers to hosts identified by IP (internet protocol) addresses and, in the case of TCP (transmission control protocol) connections, port numbers. In the context of application layer encryption, however, end-to-end is more correctly interpreted as **process-to-process**

Implementing encryption in networks

- Confidentiality between two communicating nodes is achieved by using an appropriate encryption scheme: data is encrypted at the sending node and decrypted at the receiving node.
- Encryption will also protect the traffic between the two nodes from eavesdropping to some extent.
- However, for encryption to be used effectively in networks, it is necessary to define what will be encrypted, where this takes place in the network, and the layers that are involved in a reference model.

Network security protocols

- Network security protocols are a type network protocol that ensures the security and integrity of data in transit over a network connection.
- Network security protocols define the processes and methodology to secure network data from any illegitimate attempt to review or extract the contents of data.
- Network security protocols generally implement cryptography and encryption techniques to secure the data so that it can only be decrypted with a special algorithm, logical key, mathematical formula and/or a combination of all of them.
- Some of the popular network security protocols include Secure File Transfer Protocol (SFTP), Secure Hypertext Transfer Protocol (HTTPS) and Secure Socket Layer (SSL), IPSec

ENCRYPTION PROTOCOLS

- The call for and desire for security and privacy has led to several security protocols and standards.
- Among these are: Secure Socket Layer (SSL) and Transport Layer Security (TLS) Protocols; secure IP (IPSec); Secure HTTP (S-HTTP), secure E-mail (PGP and S/MIME) and others.

IPsec

- In computing, Internet Protocol Security (**IPsec**) is a secure network protocol suite that authenticates and encrypts the packets of data on a network to provide secure encrypted communication between two computers over an Internet Protocol network.
- It is used in virtual private networks (VPNs) to encrypt transmitted packets - A virtual private network (VPN) is an encrypted connection between two or more computers.
- Based on standards developed by the Internet Engineering Task Force (IETF), IPsec ensures confidentiality, integrity, and authenticity of data communications across a public network.

IPSEC

- IPsec can protect our traffic with the following features:
  - **Confidentiality**: by encrypting our data, nobody except the sender and receiver will be able to read our data.
  - **Integrity**: we want to make sure that nobody changes the data in our packets. By calculating a hash value, the sender and receiver will be able to check if changes have been made to the packet.
  - **Authentication**: the sender and receiver will authenticate each other to make sure that we are really talking with the device we intend to.
  - **Anti-replay**: even if a packet is encrypted and authenticated, an attacker could try to capture these packets and send them again. By using sequence numbers, IPsec will not transmit any duplicate packets.

IPSEC

- Before we can protect any IP packets, we need two IPsec peers that build the IPsec tunnel.
- To establish an IPsec tunnel, we use a protocol called **IKE (Internet Key Exchange)**.
- IKE is a key management protocol used in conjunction with IPsec
- There are **two phases** to build an IPsec tunnel:
  - **IKE phase 1**
  - **IKE phase 2**

IPSEC

- In IKE phase 1, two peers will negotiate about the encryption, authentication, hashing and other protocols that they want to use and some other parameters that are required. In this phase, an **ISAKMP (Internet Security Association and Key Management Protocol)** session is established. This is also called the **ISAKMP tunnel** or **IKE phase 1** tunnel.
- The collection of parameters that the two devices will use is called a **SA (Security Association)**.



IPSEC

- The IKE phase 1 tunnel is only used for **management traffic**. We use this tunnel as a *secure method to establish the second tunnel* called the **IKE phase 2 tunnel** or **IPsec tunnel** and for management traffic like keepalives.



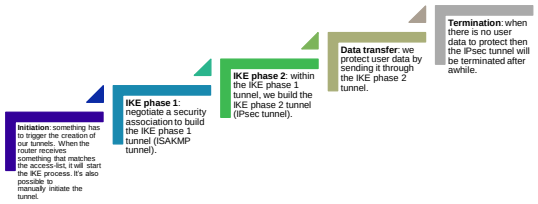
IPSEC

- IKE builds the tunnels for us but it doesn't authenticate or encrypt user data. Two other protocols are used for this:
  - **Authentication Header (AH)**—A security protocol for authenticating the source of an IP packet and verifying the integrity of its content
  - **Encapsulating Security Payload (ESP)—**A security protocol for encrypting the entire IP packet (and authenticating its content)

IPSEC Process

- IPsec provides secure *tunnels* between two peers, such as two routers.
- You define which packets are considered sensitive and should be sent through these secure tunnels, and you define the parameters which should be used to protect these sensitive packets, by specifying characteristics of these tunnels.
- Then, when the IPsec peer sees such a sensitive packet, it sets up the appropriate secure tunnel and sends the packet through the tunnel to the remote peer.

IPSEC Process



The Benefits of IPsec Technology

- When IPsec is implemented in a firewall or router, it provides strong security that can be applied to all traffic crossing the perimeter. Traffic within a company or workgroup does not incur the overhead of security-related processing.
- IPsec is below the transport layer (TCP, UDP), so is transparent to applications.
- There is no need to change software on a user or server system when IPsec is implemented in the firewall or router. Even if IPsec is implemented in end systems, upper layer software, including applications, is not affected.

Internet Key Exchange (IKE)

- Internet Key Exchange (IKE) is a secure key management protocol that is used to set up a secure, authenticated communications channel between two devices.
- The goal of the *Internet Key Exchange* (IKE) is for both sides to independently produce the same symmetrical key.
- This key then encrypts and decrypts the regular IP packets used in the bulk transfer of data between VPN peers.
- IKE builds the VPN tunnel by authenticating both sides and reaching an agreement on methods of encryption and integrity. The outcome of an IKE negotiation is a *Security Association* (SA).

Internet Key Exchange (IKE)

- IKE does the following:
  - Negotiates and manages IKE and IPsec parameters
  - Authenticates secure key exchange
  - Provides mutual peer authentication by means of shared secrets (not passwords) and public keys
  - Provides identity protection (in main mode)
  - Employs Diffie-Hellman methods and is optional in IPsec (the shared keys can be entered manually at the endpoints).

Interaction Between IKE and IPSec

- IPsec can establish a VPN in either of the following way:
- Internet Key Exchange (IKE) protocol— IPsec supports automated generation and negotiation of keys and security associations using the IKE protocol.
- Using IKE to negotiate VPNs between two endpoints provides more security than the manual key exchange.
- Manual key exchange—IPsec supports using and exchanging of keys manually (example: phone or email) on both sides to establish VPN.

Secure Socket Layer (SSL)

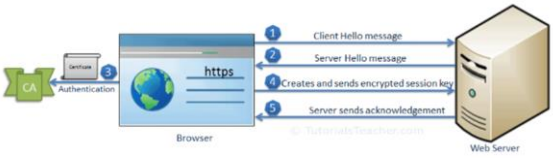
- Secure Sockets Layer is a standard security mechanism used for preserving a secure internet connection by safeguarding different sensitive data which is being sent and receives between 2 systems; which also helps in averting cybercriminals from reading as well as modifying personal data, packets or details in the network.
- SSL and the superseding Transport Layer Security (TLS) protocols resolve this issue by encrypting data when it leaves a web server and decrypting it when it arrives at its destination.
- SSL is used for websites, email, remote login, and much more.
- A Website which has a SSL certificate will contain a small padlock before the website's URL. This means that the site is secured to conduct transactions.

Secure Socket Layer (SSL)

- The SSL protocol provides data encryption
- SSL works through the use of public key cryptography. Public key cryptography uses two keys – a private key and a public key – to transmit secure data between the webserver and the browser.
- It also provides server and client authentication using X.509 certificates.
- SSL begins with a handshake during which the server is authenticated to the client using X.509 certificates. Also, the client can optionally be authenticated to the server.
- During the handshake, security session parameters, such as cryptographic algorithms, are negotiated and session keys are created.
- After the handshake, the data is protected during transmission with data origin authentication and optional encryption using the session keys.

SSL HANDSHAKE

- The SSL handshake is an asymmetric cryptography which allows the browser to verify the web server, get the public key and establish a secure connection before the beginning of the actual data transfer.



SSL HANDSHAKE

1. The client sends a "client hello" message. This includes the client's SSL version number, cipher settings, session-specific data and other information that the server needs to communicate with the client using SSL.
2. The server responds with a "server hello" message. This includes the server's SSL version number, cipher settings, session-specific data, an SSL certificate with a public key and other information that the client needs to communicate with the server over SSL.
3. The client verifies the server's SSL certificate from CA (Certificate Authority) and authenticates the server. If the authentication fails, then the client refuses the SSL connection and throws an exception. If the authentication succeeds, then proceed to step 4.
4. The client creates a session key, encrypts it with the server's public key and sends it to the server. If the server has requested client authentication (mostly in server to server communication), then the client sends his own certificate to the server.
5. The server decrypts the session key with its private key and sends the acknowledgement to the client encrypted with the session key.
6. Thus, at the end of the SSL handshake, both the client and the server have a valid session key which they will use to encrypt or decrypt actual data. The public key and the private key will not be used any more after this

ADVANTAGES OF SSL

1. Security -The major goal of SSL certificate is to encrypt information so that it can only be read by intended recipients. SSL makes it ideal for protecting sensitive information such as the User IDs, passwords and credit card numbers.

2. Authentication - SSL ensures that whatever the information that is present on your site reach to the correct server. For achieving this SSL uses a protection known as the Server certificate. This Server certificate makes sure that the SSL certificate provider is trusted by acting as a middle person between the browsers and SSL servers.

3. Reliability- Whenever a SSL certificate is used on a site, it provides verification. Hence, when visitors visit a site, it provides a sense of trust that this site is legitimate and not fake

4. Online Payments- All the payment card industries require sites to have a SSL certificate with at least 128 bit encryption to accept payments. Without a proper SSL certificate sites won't be able to accept payments from credit cards.

DISADVANTAGES OF SSL

1. **Performance -**When a SSL certificate is used on a site, the speed of transactions will get drastically reduced. This happens because of the encryption and decryption of data before it can be used.
2. **Expiry-** Occasionally the SSL certificate needs to be renewed. If it is not renewed time to time, there will be popup message stating that the SSL certificate has been expired which means that the site is not secure anymore. Hence, the customers can loose trust in conducting transactions.

S-HTTP (Secure HTTP)

- S-HTTP (Secure HTTP) is an extension to the Hypertext Transfer Protocol (HTTP) that allows the secure exchange of files on the World Wide Web.
- Each S-HTTP file is either encrypted, contains a digital certificate, or both.
- For a given document, S-HTTP is an alternative to another well-known security protocol, Secure Sockets Layer (SSL). A major difference is that S-HTTP allows the client to send a certificate to authenticate the user whereas, using SSL, only the server can be authenticated.
- S-HTTP is more likely to be used in situations where the server represents a bank and requires authentication from the user that is more secure than a userid and password.
- The principal motivations for HTTPS are authentication of the accessed website, and protection of the privacy and integrity of the exchanged data while in transit.

S-HTTP

- It protects against man-in-the-middle attacks, and the bidirectional encryption of communications between a client and server protects the communications against eavesdropping and tampering.
- The authentication aspect of HTTPS requires a trusted third party to sign server-side digital certificates.
- S-HTTP was designed for use in browser applications, so it cannot be used to secure non-browser-based Internet applications.

S/MIME

- Secure/Multipurpose Internet Mail Extension (S/MIME) is an email encryption and signing industry standard widely used by corporations to enhance email security. S/MIME is compatible with most enterprise email clients.
- S/MIME (Secure/Multipurpose internet Mail Extensions) is a widely accepted protocol for sending digitally signed and encrypted messages.
- S/MIME in Exchange Online provides the following services for email messages:
  - **Encryption**: Protects the content of email messages.
  - **Digital signatures**: Verifies the identity of the sender of an email message.
- In simple terms, S/MIME encrypts and digitally signs emails to ensure that the email is authenticated and its contents have not been altered in any way.

S/MIME

- S/MIME is based on asymmetric cryptography to protect your emails from unwanted access. It also allows you to digitally sign your emails to verify you as the legitimate sender of the message, making it an effective weapon against many phishing attacks out there –You wont need a pen to do this.
- Every time to you create and sign an email, your private key applies your unique Digital Signature into your message. When your recipient opens your email, your public key is used to verify the signature.
- This ensures your recipient that the emails really came from you.
- Signing emails authenticates your identity in an age where phishing attacks have already become so clever and it has become increasingly difficult to identify spoofed emails.

S/MIME

- Some of the popular desktop and mobile email clients that support S/MIME include:
  - Apple Mail
  - CipherMail (for Android mobile devices)
  - Gmail (G Suite Enterprise and G Suite for Education)
  - IBM Notes
  - iPhone iOS Mail
  - MailMate
  - Microsoft Outlook and Outlook on the Web (formerly Outlook Web App)
  - Mozilla Thunderbird

S/MIME

- Although S/MIME is supported by many email clients, it has not been widely implemented. Historically, it has been used by primarily by highly confidential government types and technology geeks because it was too cumbersome to for general users to implement.
- Part of this is due to the fact that both the sender and the recipient need to have S/MIME certificates to encrypt and decrypt one another's messages using public and private keys.

Benefits of S/MIME

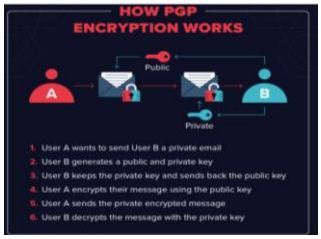
- Through encryption and the use of digital signatures, S/MIME helps to protect users from email-related cyber attacks such as phishing, spear-phishing, CEO fraud, "whaling," and more.
- S/MIME email signing certificates prevent malicious users from "eavesdropping" on your email communications — even if they are actively "listening," your plaintext communications will still be indecipherable because of the encryption.

PGP

- Pretty Good Privacy (PGP) is an encryption system used for both sending encrypted emails and encrypting sensitive files. Since its invention back in 1991, PGP has become the de facto standard for email security.
- The popularity of PGP is based on two factors.
  - The first is that the system was originally available as freeware, and so spread rapidly among users who wanted an extra level of security for their email messages.
  - The second is that since PGP uses both symmetric encryption and public-key encryption, it allows users who have never met to send encrypted messages to each other without exchanging private encryption keys.
- PGP encrypts and signs the message, including attachments, as a whole.
- An advantage to this scheme is that the message structure, like attachment metadata, is not leaked to someone who intercepts the encrypted message.

PGP

- There are, essentially, three main uses of PGP:
  - Sending and receiving encrypted emails.
  - Verifying the identity of the person who has sent you this message.
  - Encrypting files stored on your devices or in the cloud.
- Of these three uses, the first – sending secure email – is by far the dominant application of PGP.



Advantages and Disadvantages of PGP

Advantages	Disadvantages
• Extremely Secure • Open PGP is free to use • Improves cloud security	• Not user friendly • Requires software • No annonity

Take home: To Read

- **Difference between PGP and S/MIME**

Cryptanalysis

- Cryptanalysis is used as a method of decrypting a cipher text into plaintext.
- An unauthorized person tries to decrypt the message by eavesdropping on the unsecured channel. It is also known as code-breaking.
- This person is not bound by any rules i.e a black hat hacker